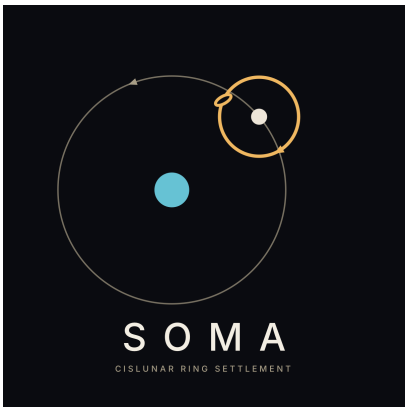


Soma: CisLunar Rings

A cislunar ring settlement design study

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3 November 2026

1 Summary

Soma is a speculative design study for a cislunar ring settlement: six rings, each 109 km around (34.7 km across), on one shared spine, housing about 3.3 million people, parked in a stable loop around the Moon. Each ring is built mostly from lunar material by self-assembling bay factories, at an estimated $87k–661k of build cost per resident. All figures are order-of-magnitude estimates for a concept, not engineering results.

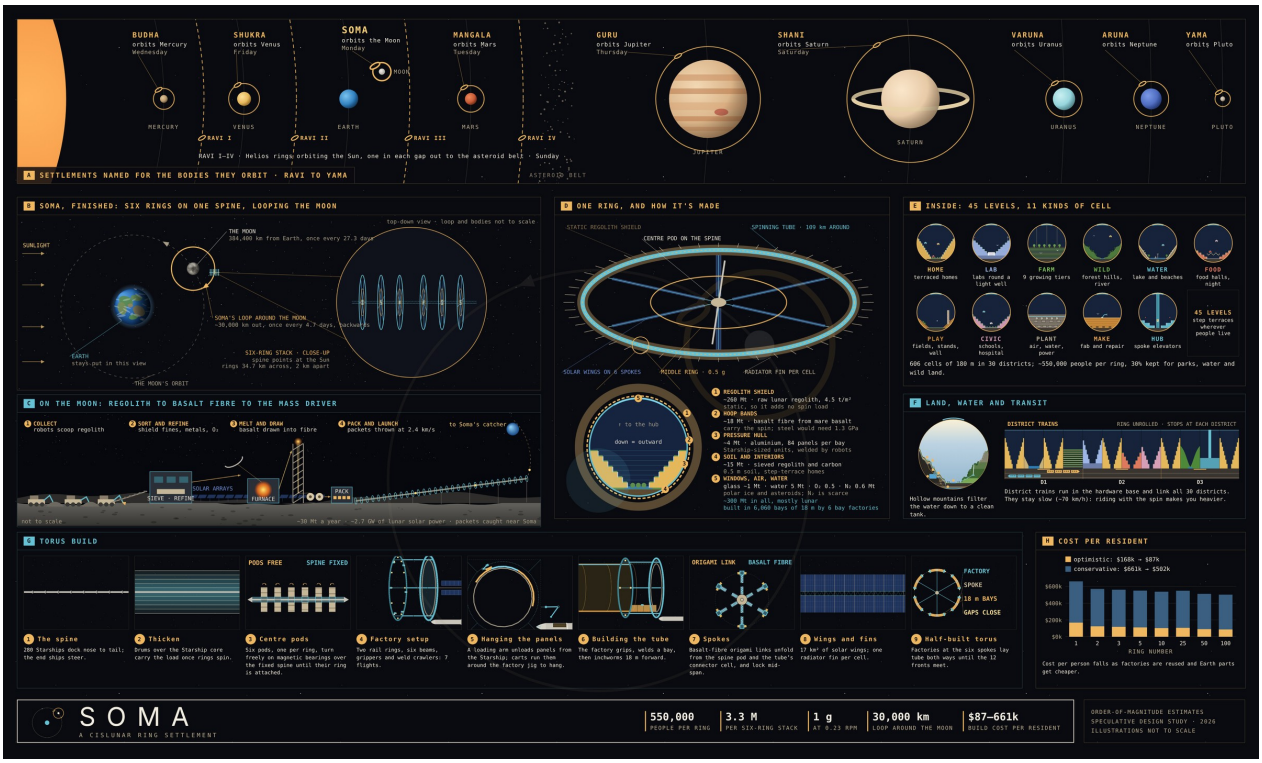


Figure 1: The Soma poster (60 × 36 in). An animated version, in which every tile plays its build step, is in the repository.

2 The design

Each ring is a 150 m-wide tube bent into a 17.35 km-radius torus that spins at 0.23 rpm for 1 g at the floor:

$$a = \omega^2 R, \qquad \omega = \sqrt{\frac{9.81\ \mathrm{m\,s^{-2}}}{17\,350\ \mathrm{m}}} \approx 0.0238\ \mathrm{rad\,s^{-1}} \approx 0.23\ \mathrm{rpm}, \qquad v = \omega R \approx 412\ \mathrm{m\,s^{-1}}. \quad (1)$$

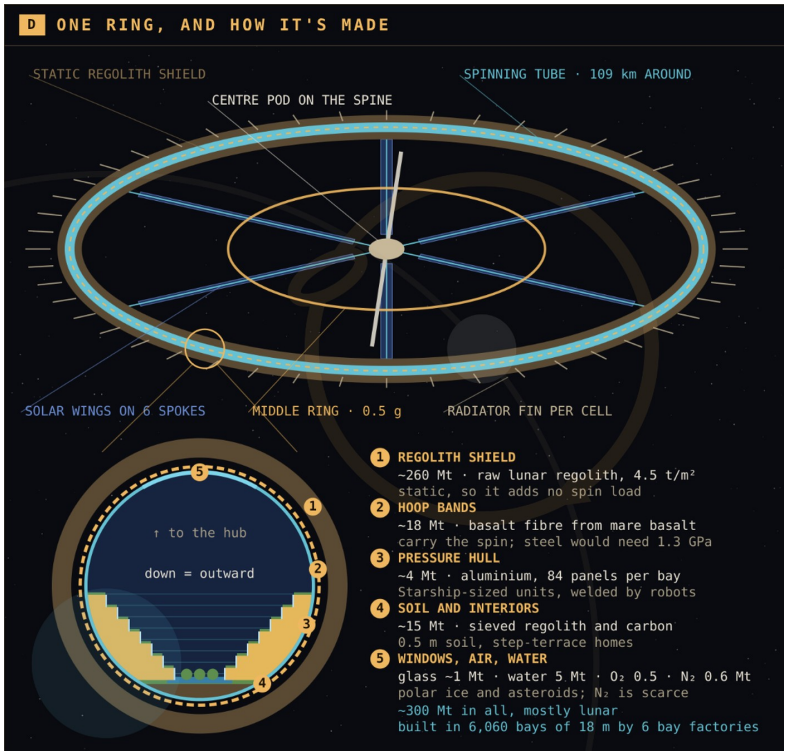


Figure 2: One ring and how it is made: shield, hoop bands, hull, soil and terraced interiors.

- **Tube and shield.** Basalt-fibre hoop bands carry the spin. The hoop stress from a ring's own mass is $\sigma = \rho v^2$, so steel would need 1.3 GPa and is too heavy. A static shell of raw regolith (∼260 Mt) shields against radiation and adds no spin load.
- **Cells.** The tube is split into 606 cells of 180 m, grouped into 30 districts. Each cell has 45 levels of 3.33 m and is one of 11 types (home, lab, farm, wild, water, food and night, play, civic, plant, make, hub).
- **Step terraces.** Every cell where people live or work steps back from a central canyon, so each level has open sky and its own terrace.
- **Spine.** 280 Starships dock nose to tail into a 14 km core, later wrapped in five concentric drums out to $r = 100$ m (∼1.4 Mt).
- **Centre pods.** Six pods, one per ring, turn on magnetic bearings around the fixed spine.
- **Spokes.** Origami-style basalt-fibre links unfold from both the pod and the tube's connector cell and lock mid-span. Spin pulls $\sigma = \rho v^2/2 \approx 0.23$ GPa on a spoke; basalt fibre holds ∼1.5 GPa.
- **Stack.** Six rings in three counter-rotating pairs, 2 km apart, so the stack has no net spin and can be steered like a gyroscope to face the Sun.
- **Power and heat.** Solar wings on the six spokes (∼6 GW) face the Sun at all times because the spin axis points at it. About 650 radiator fins, one per cell, shed heat edge-on.

- **Transit.** District trains run through the hardware base at ~70 km/h. They stay slow because riding with the spin makes you feel heavier.

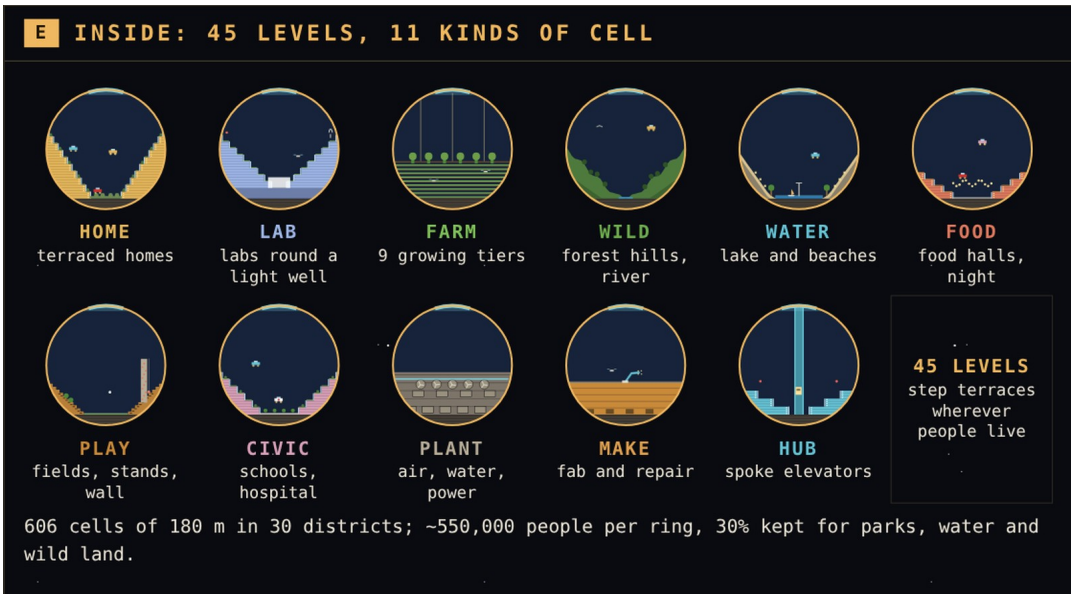


Figure 3: The 11 cell types in cross-section, with step terraces wherever people live.



Figure 4: Land, water and transit: a filter mountain and district trains in the hardware base.

3 How it is built

The build runs in nine steps, starting from the spine and ending when the tube closes into a ring. The full construction film, *How Soma is built* (5 min 57 s), is in the repository under `videos/soma-construction-film.mp4`.

1. **The spine.** Cargo Starships unload, then dock nose to tail along the spin axis; the ships at each end keep their engines to steer.
2. **Thicken.** Concentric drums are wrapped over the Starship core so it can carry Sun-tracking loads once the rings spin.
3. **Centre pods.** Six pods are built over the fixed spine one by one. They turn freely on magnetic bearings; real spin waits until the ring is attached.
4. **Factory setup.** Seven Starship flights bring two rail rings, six beams, grippers and weld crawlers for each bay factory.

5. **Hanging the panels.** A loading arm unloads hull panels from a Starship; carts run them around the factory jig to hang.
6. **Building the tube.** The factory grips the tube, welds and leak-tests one 18 m bay, then inchworms forward.
7. **Spokes.** Origami links unfold from the pod and the tube's connector cell and lock mid-span, in balanced pairs.
8. **Wings and fins.** Wing robots unroll 17 km^2 of lunar-made solar cells along the spokes; carriers fit one radiator fin per cell.
9. **Half-built torus to full ring.** Six factories build tube both ways from the spokes; the 12 fronts meet after 6,060 bays and the ring closes.

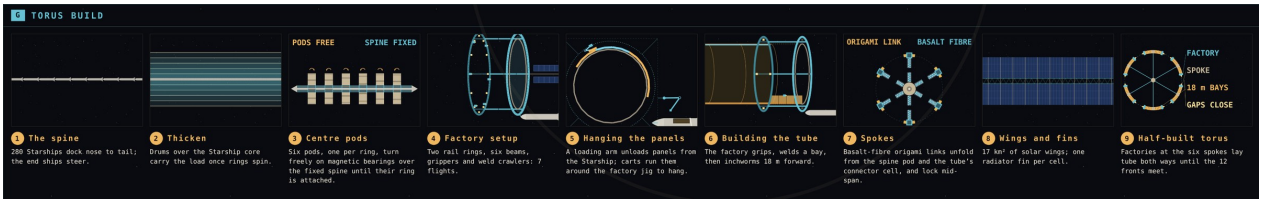


Figure 5: The nine build steps, from the spine to the half-built torus.

On the Moon, robots collect regolith, plants sort it and draw basalt into fibre, and a mass driver throws packets at 2.4 km/s to a catcher near Soma (~30 Mt a year).

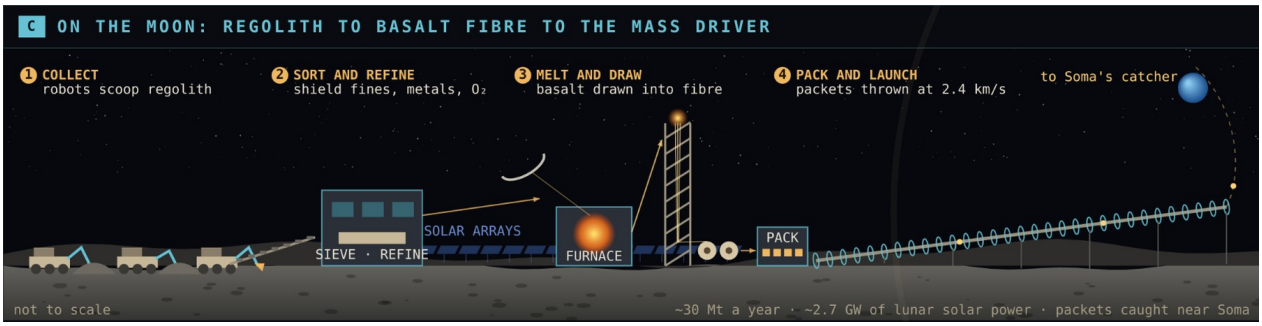


Figure 6: On the Moon: regolith to basalt fibre to the mass driver.

4 Materials and where they come from

Over 99% of a ring's mass is lunar: about 320 Mt once interior decks are counted (the ~300 Mt used elsewhere leaves them out). Earth supplies the precise fraction of a per cent: electronics, machines, batteries, biology and, unless asteroids or polar ice deliver it, nitrogen.

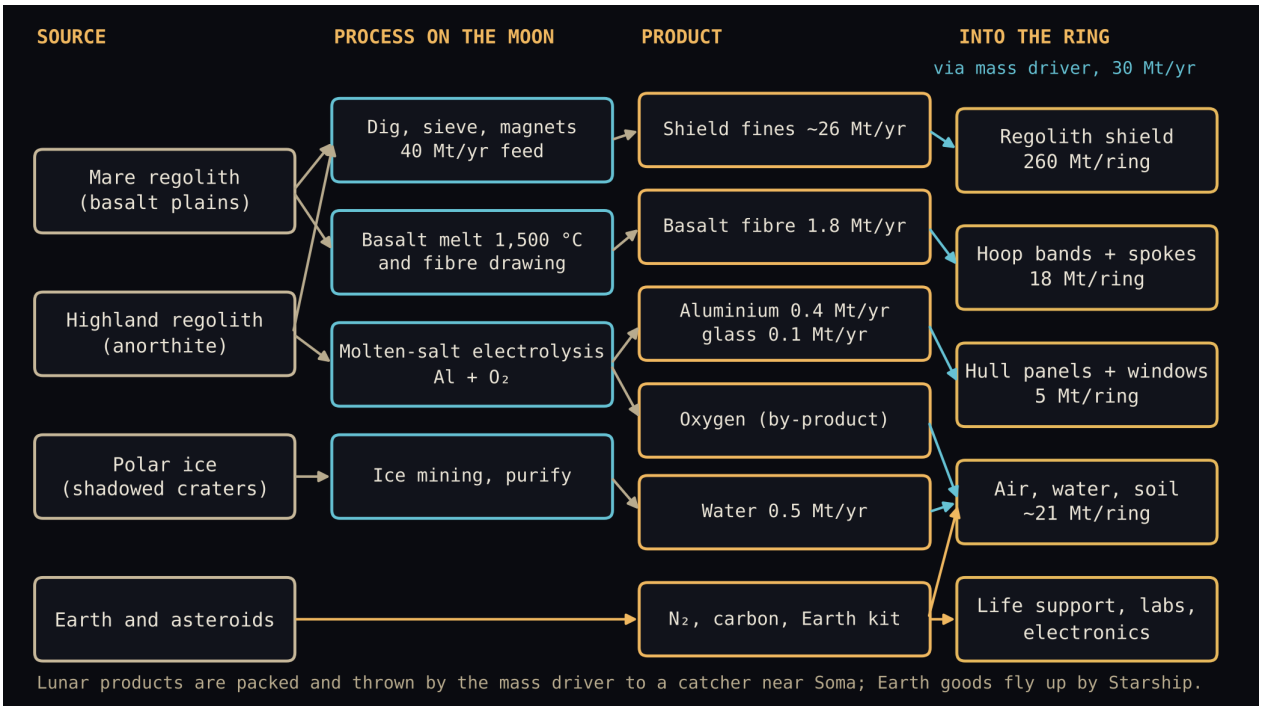


Figure 7: Material flow from lunar and Earth sources to the ring. Lunar products are packed and thrown by the mass driver; Earth goods fly up by Starship.

Material	Per ring	Source	Process
Regolith shield	260 Mt	Mare and highland surface regolith	Dug, sieved, packed into bags; never spins
Basalt fibre (hoops, spokes)	18 Mt	Mare basalt	Melted at 1,500 °C, drawn through bushings, wound into bands
Aluminium (hull, decks)	4 Mt	Highland anorthite	Molten-salt electrolysis; oxygen is a by-product
Glass (windows)	1 Mt	Regolith silica	Melted and float-formed on the Moon
Decks and frames	15 Mt	Aluminium, basalt composite, sintered regolith	Pressed panels and printed blocks
Soil	15 Mt	Sieved regolith + carbon	Carbon from C-type asteroids or Earth; inoculated with soil biology from Earth
Water	5 Mt	Polar ice in shadowed craters	Mined, melted, purified
Oxygen	0.5 Mt	Regolith oxides	By-product of aluminium and oxygen extraction
Nitrogen	0.3–0.6 Mt	Earth today; asteroids or polar volatiles later	Shipped as liquid; the main open supply question
Earth high-tech kit	50,000 t	Earth	Electronics, life-support machines, lab gear, seeds and microbes
Batteries	37,500 t	Earth	6 GWh of storage for spin-up, eclipses and peaks

Table 1: Bill of materials for one ring, with sources.

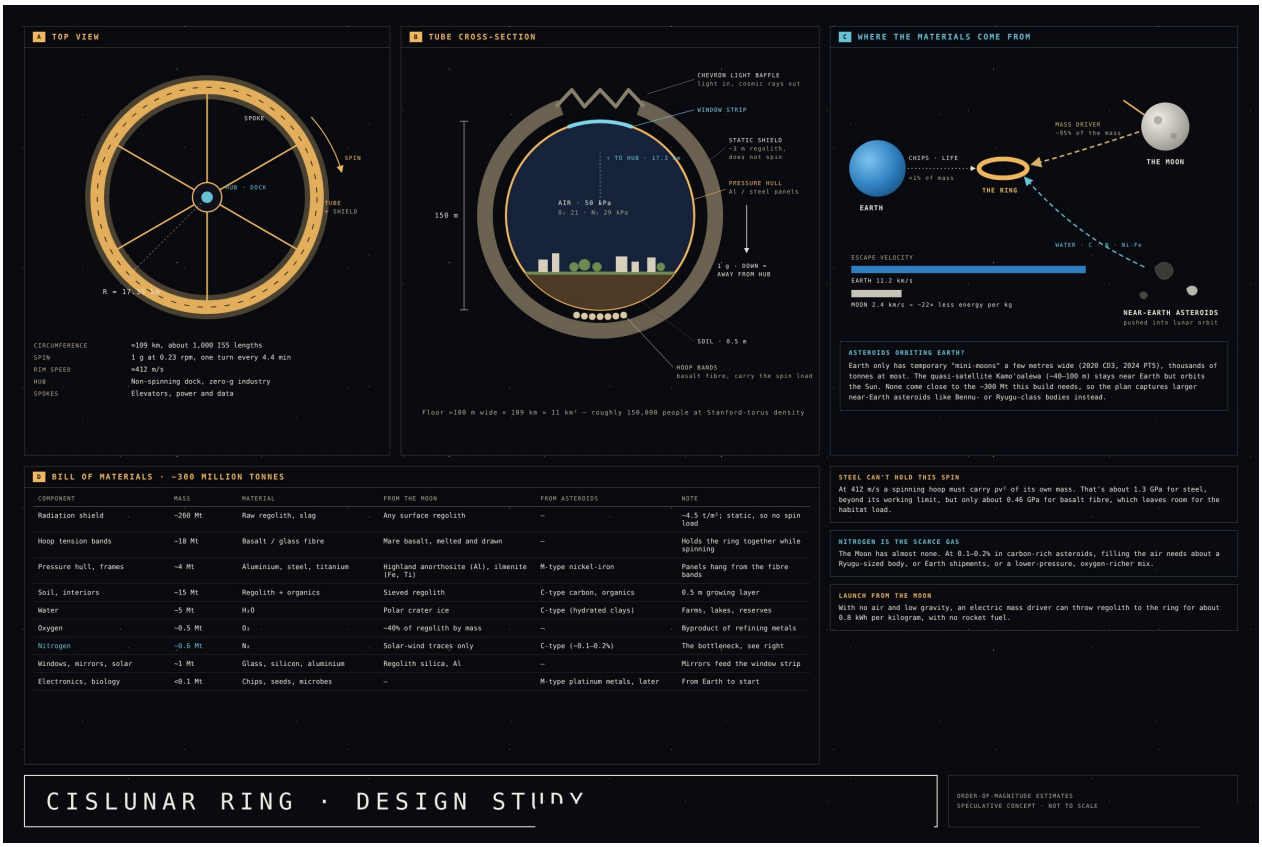


Figure 8: Design board: the ring, its 150 m tube, the static regolith shield and the bill of materials.

5 Materials and cost by cell type

A ring has 606 cells of 180 m, in 11 types. Every cell carries the same shell (∼465 kt of shield, hoop bands, hull and glass); what differs is the interior and the Earth-made kit inside it.

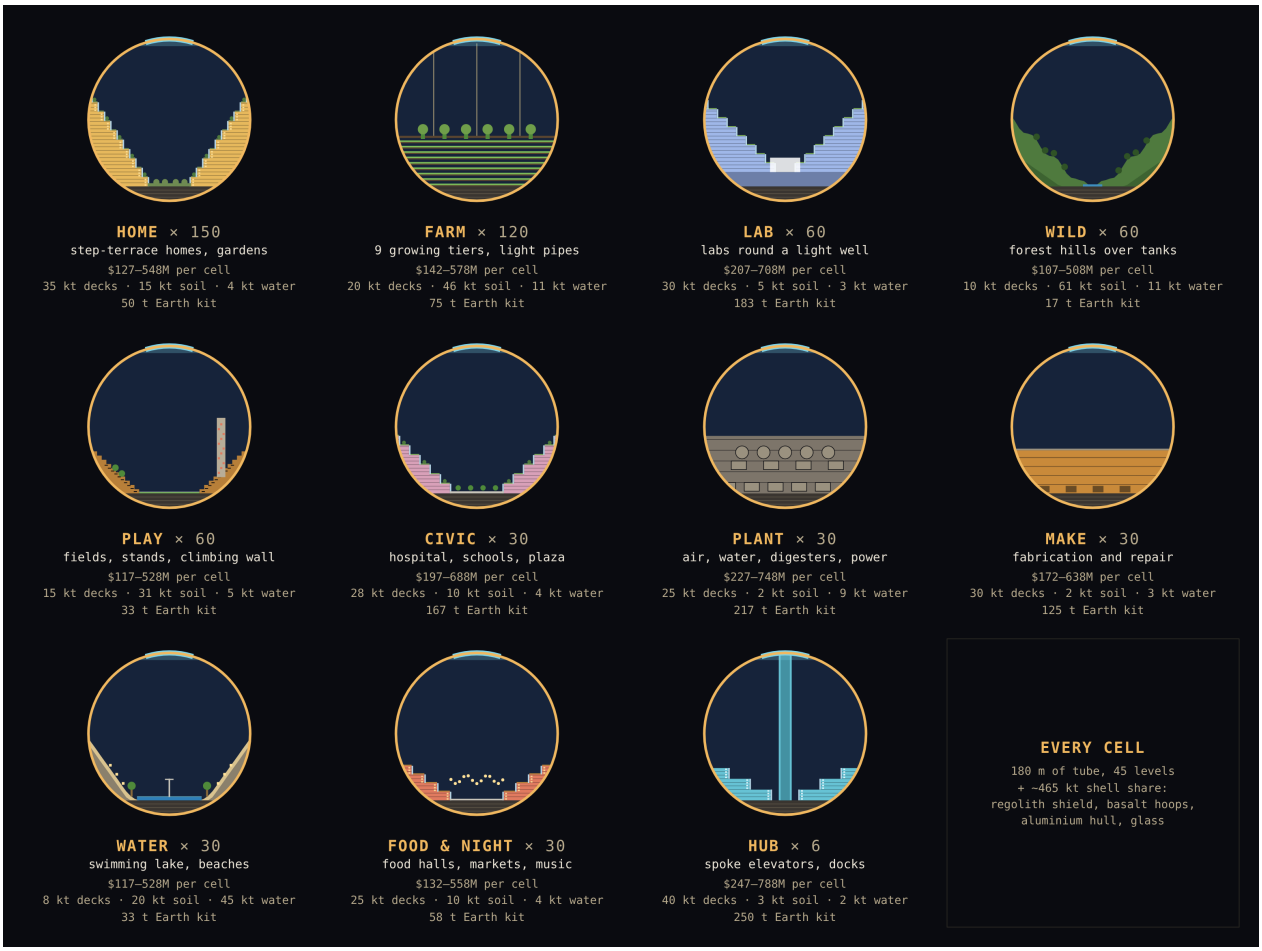


Figure 9: The 11 cell types, with their count per ring, build cost per cell, lunar interior mass and Earth kit.

How the cost is split. The ring's standalone cost ($89B optimistic, $356B conservative) is the sum of six items from the cost model: share of the lunar industry, Earth high-tech kit, nitrogen, the power system, robots and bay factories, and design and start-up. Five of these are shared equally by all cells ($97M a cell optimistic, $488M conservative). The Earth kit is split by what each type needs: a hub cell carries 250 t of elevators, docks and controls, a wild cell only 17 t of pumps and sensors.

Cell type	Cells	Decks kt	Soil kt	Water kt	Earth kit t	Cost per cell $M	All cells of this type $B
Home	150	35	15	4	50	127–548	19.1–82.3
Farm	120	20	46	11	75	142–578	17.1–69.4
Lab	60	30	5	3	183	207–708	12.4–42.5
Wild	60	10	61	11	17	107–508	6.4–30.5
Play	60	15	31	5	33	117–528	7.0–31.7
Civic	30	28	10	4	167	197–688	5.9–20.7
Plant	30	25	2	9	217	227–748	6.8–22.5
Make	30	30	2	3	125	172–638	5.2–19.2
Water	30	8	20	45	33	117–528	3.5–15.9
Food & night	30	25	10	4	58	132–558	4.0–16.8
Hub	6	40	3	2	250	247–788	1.5–4.7
Ring	606	14.7 Mt	15.0 Mt	5.0 Mt	50,000		**89–356**

Table 2: Lunar interior mass, Earth kit and build cost for each cell type. Shell mass is equal for all cells and not shown. Ranges are optimistic–conservative.

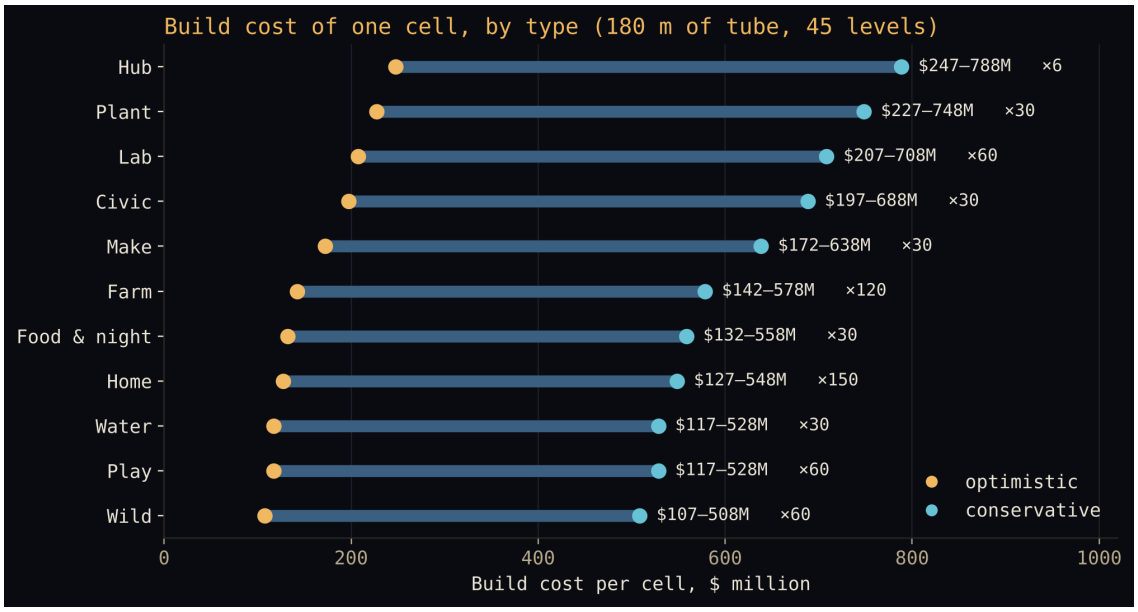


Figure 10: Build cost of one cell, by type. The spread is set mostly by nitrogen and power prices, not by the cell's own fit-out.

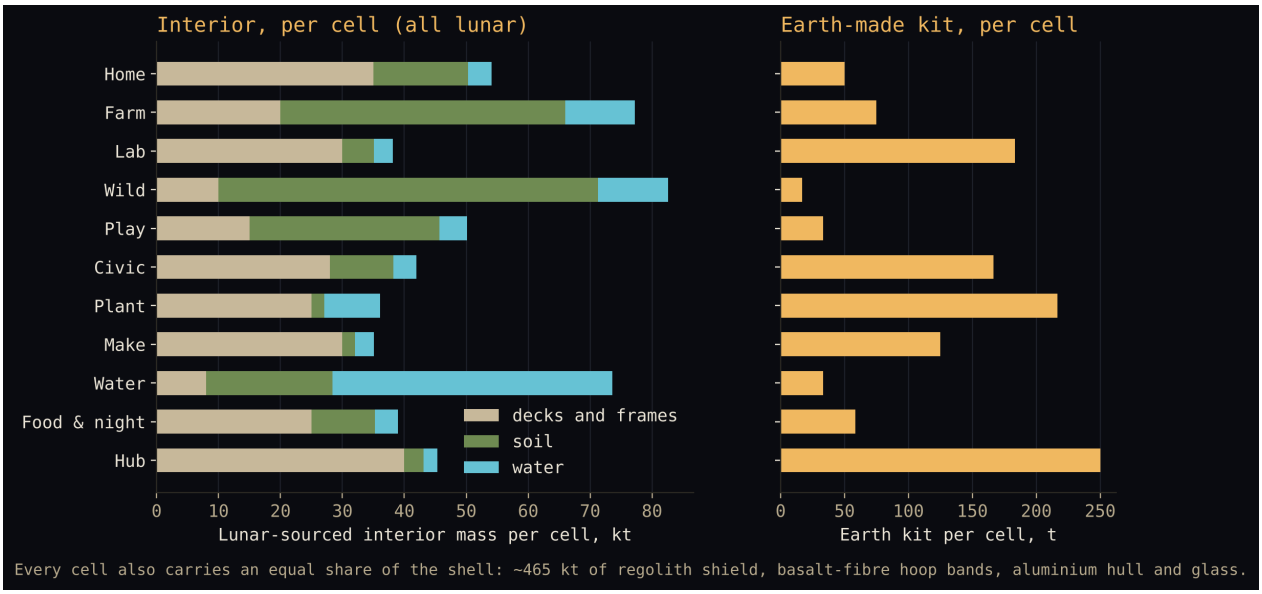


Figure 11: What goes inside each cell type: lunar decks, soil and water (left) and Earth-made kit (right).

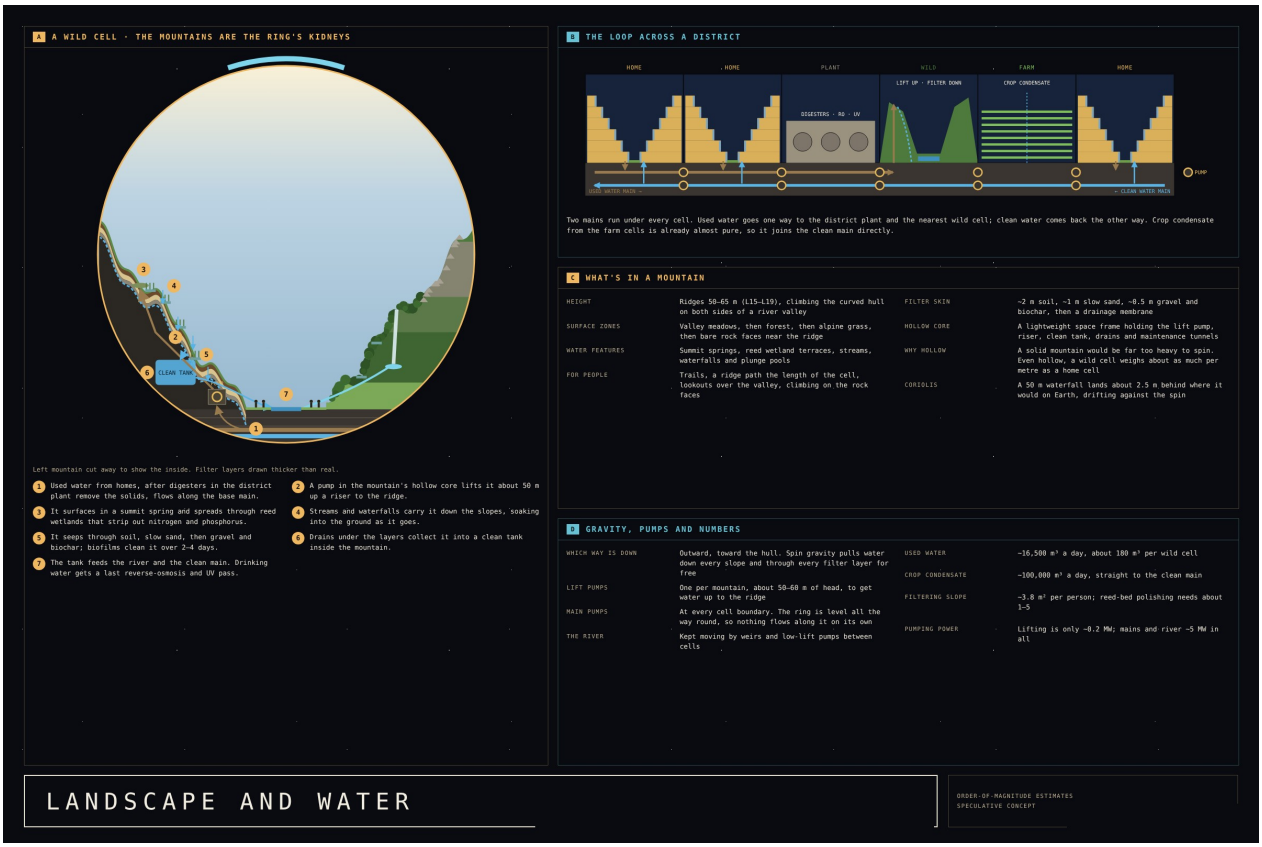


Figure 12: Design board: hollow filter mountains and the district water loop, which set the soil and water mass of the wild, water and farm cells.

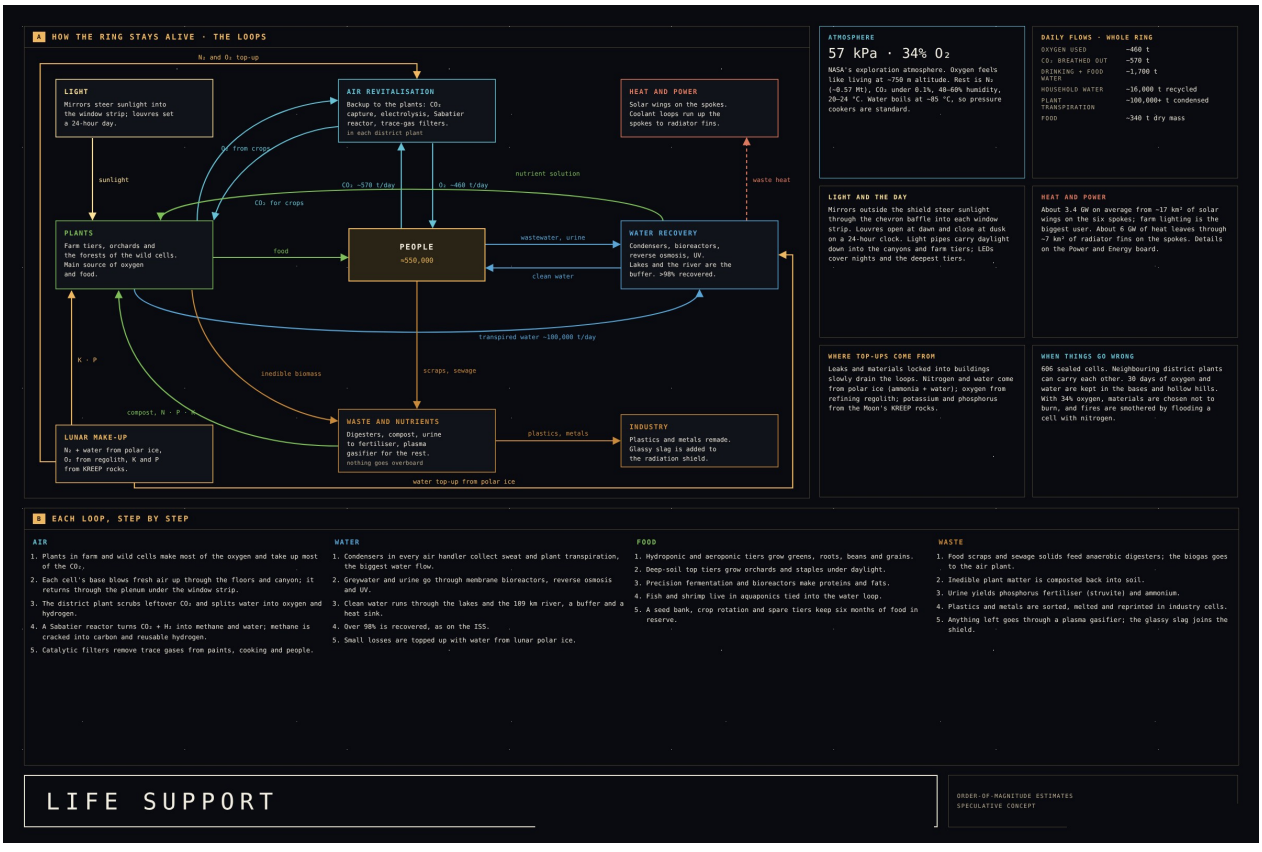


Figure 13: Design board: air, water, food and waste loops. Most of the Earth kit in plant and farm cells is this machinery.

6 The lunar base

One mass-driver line delivers 30 Mt a year, enough for one ring about every 10 years. Rings faster than that need more lines: ten lines give a ring a year. The base sits on a polar ridge with near-constant sunlight, next to a permanently shadowed crater for ice and within haul range of mare and highland regolith.

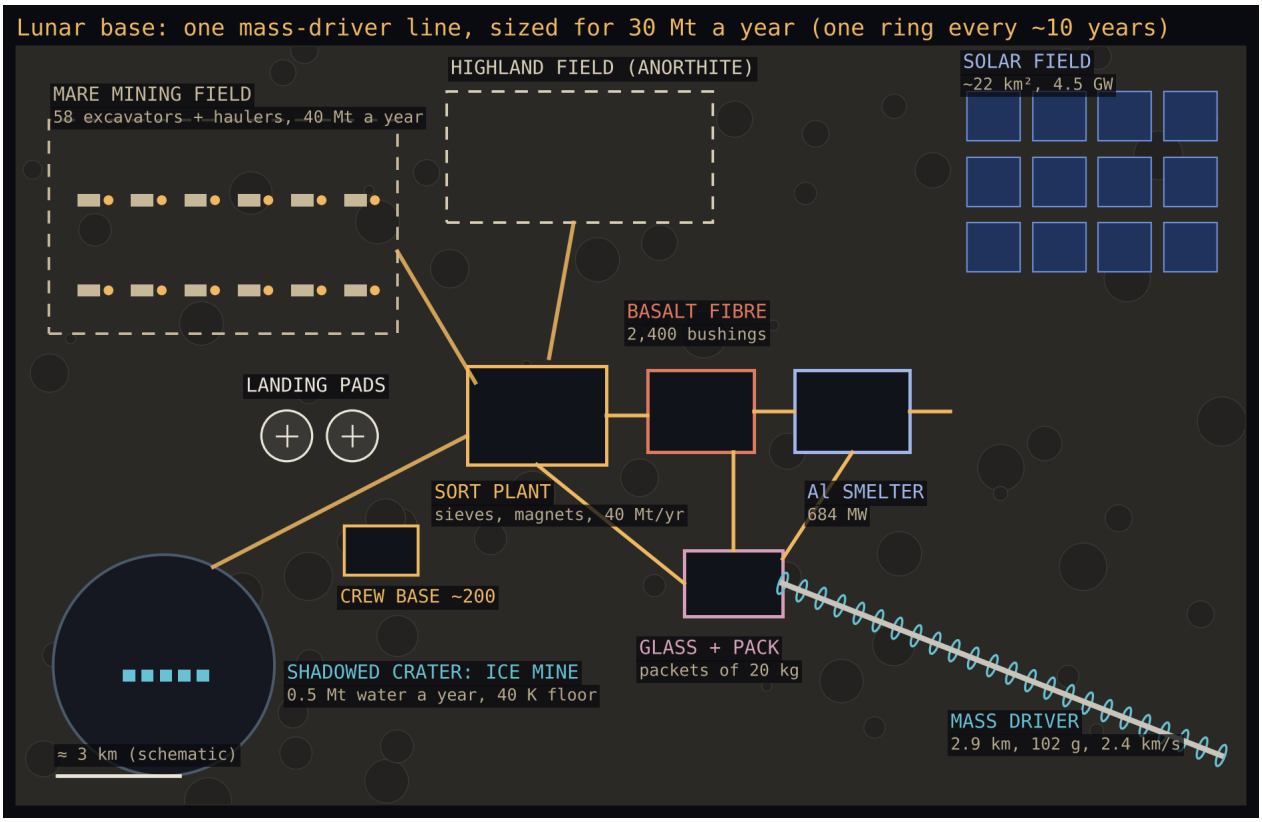


Figure 14: Lunar base site plan (schematic): mining fields, sort plant, fibre works, smelter, glass and packing, ice mine, solar field, landing pads and the mass driver.

Unit	Spec	Power
Mining fleet	58 bucket-wheel excavators (100 t/h each, 7,000 h/yr) plus haulers; 40 Mt/yr of regolith dug from mare and highland fields	91 MW
Sort plant	Sieves, magnetic and electrostatic separation; about 75% of the feed is shipped, the rest is tailings	
Basalt fibre works	Electric furnaces at 1,500 °C feeding 2,400 platinum–rhodium bushings (100 kg/h each); 1.8 Mt/yr of continuous fibre	57 MW
Aluminium smelter	Molten-salt electrolysis of anorthite at ~15 MWh/t; 0.4 Mt/yr of aluminium and oxygen as a by-product	684 MW
Glass line	Regolith melt to window glass; 0.1 Mt/yr	6 MW
Ice mine	Rovers and heated tents in a permanently shadowed crater (40 K floor); 0.5 Mt/yr of water	57 MW
Packer and mass driver	20 kg packets, 48 a second; 2.88 km coilgun at 102 g to 2.4 km/s; 30 Mt/yr	3,422 MW
Crew base, pads, workshops	~200 crew, 2 landing pads of sintered regolith, robot repair shops	150 MW
Solar field	~22 km^2 of arrays at ~200 W/m^2 on a polar ridge with near-constant sun	**4.5 GW**

Table 3: Lunar base units for one mass-driver line, with their electrical demand.

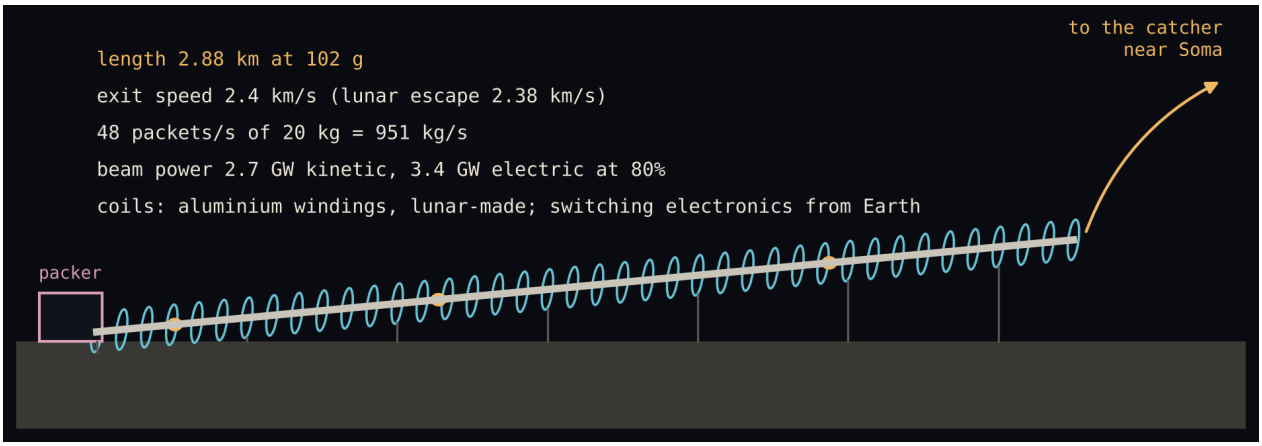


Figure 15: The mass driver: a coilgun that throws 20 kg packets just above lunar escape speed toward a catcher near Soma.

The launch speed sets the length and the power:

$$L = \frac{v^2}{2a} = \frac{(2400\ \mathrm{m\,s^{-1}})^2}{2 \times 1000\ \mathrm{m\,s^{-2}}} \approx 2.9\ \mathrm{km}, \qquad P = \tfrac{1}{2}\,\dot{m}\,v^2 = \tfrac{1}{2} \times 951\ \mathrm{kg\,s^{-1}} \times (2400\ \mathrm{m\,s^{-1}})^2 \approx 2.7\ \mathrm{GW}. \tag{2}$$

Building the base. Earth lands a 20,000 t seed factory: excavators, a pilot sort plant, a first furnace and bushings, a small smelter, 200 MW of arrays and a crew base. The base then grows mostly from its own output, with ∼250,000 t of Earth-made parts over its life (motors, power electronics, bushings, sensors). The programme costs $0.5–1.9T, shared by 100 rings ($5–19B a ring).

7 Starship launches

Each delivery beyond low Earth orbit needs tanker flights to refuel the ship first: about 6 per 100 t delivered to Soma's orbit and 10–15 per 100 t landed on the Moon.

Stage	Payload	Launches
Once, shared by all rings (to the lunar surface: 100 t a landing, 10–15 tankers each)		
Lunar seed factory	20,000 t	2,200–3,200
Lunar full build-out (Earth-made parts)	250,000 t	27,500–40,000
Each ring (to Soma's orbit: 100 t a delivery, ∼6 tankers each)		
Spine share (280 Starships per 6-ring stack)	47 ships	327
Bay factories (6 × 7 flights)	42 loads	294
Robot fleet	5,000 t	350
Earth high-tech kit	50,000 t	3,500
Batteries	37,500 t	2,625
Subtotal, without nitrogen		**7,096**
Nitrogen from Earth	0.3–0.6 Mt	21,000–42,000
Per ring, all in		**28,096–49,096**

Table 4: Starship launches by stage, tanker flights included.

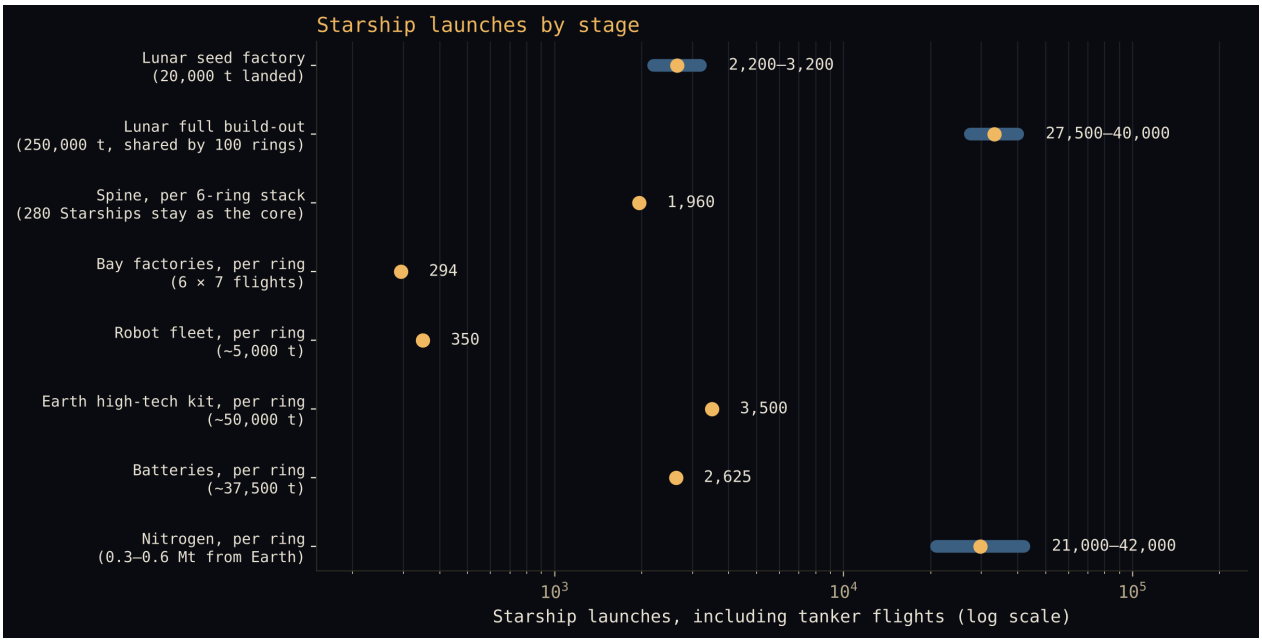


Figure 16: Launches by stage on a log scale. Nitrogen dominates: sourcing it from asteroids or lunar polar volatiles cuts a ring's launches from ∼28,000–49,000 to ∼7,100.



Figure 17: Design board: Starship-sized hull units, 84 panels per bay, and robotic assembly of one bay.

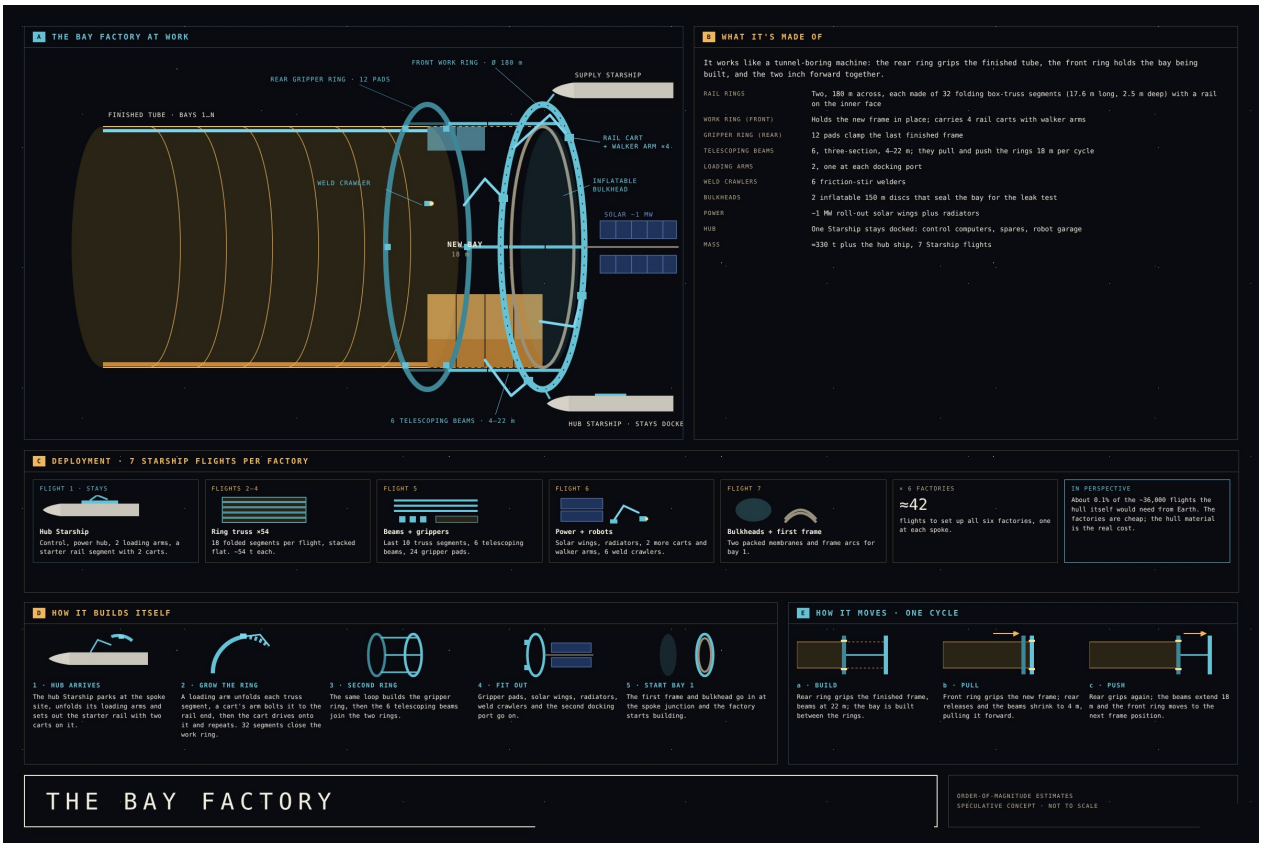


Figure 18: Design board: the bay factory, its 7 Starship flights and the inchworm cycle.

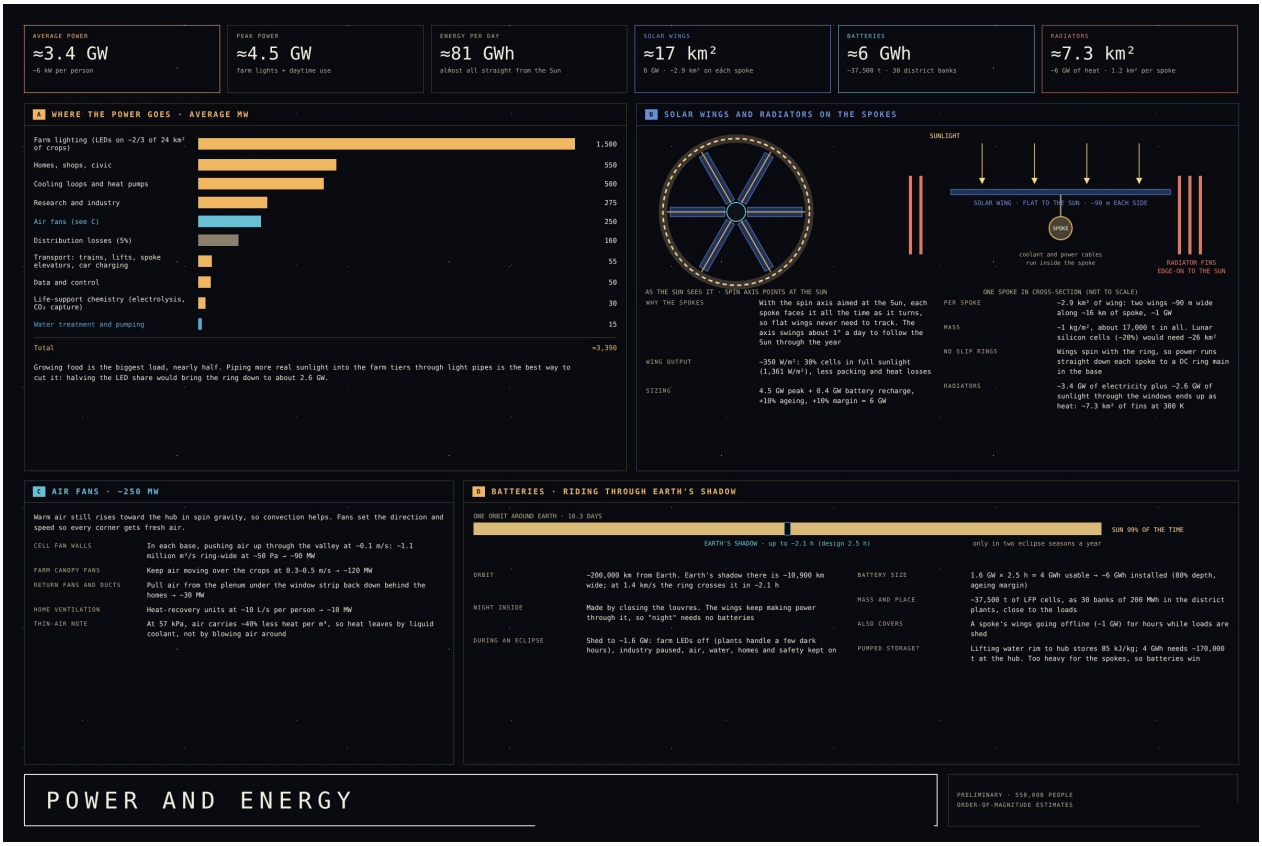


Figure 19: Design board: spoke solar wings, batteries and loads, which set the battery tonnage in the launch count.

8 Orbit

Soma sits in a lunar distant retrograde orbit about 30,000 km from the Moon, looping it backwards every 4.7 days while the Moon carries it around Earth. In a simplified Earth–Moon–Sun model it stayed bound for 20 years with no corrections, even with a 2 m/s kick each time a ring was added. Larger loops (40,000–70,000 km) drifted off within 2–4 years. The Soma logo is drawn from this orbit.

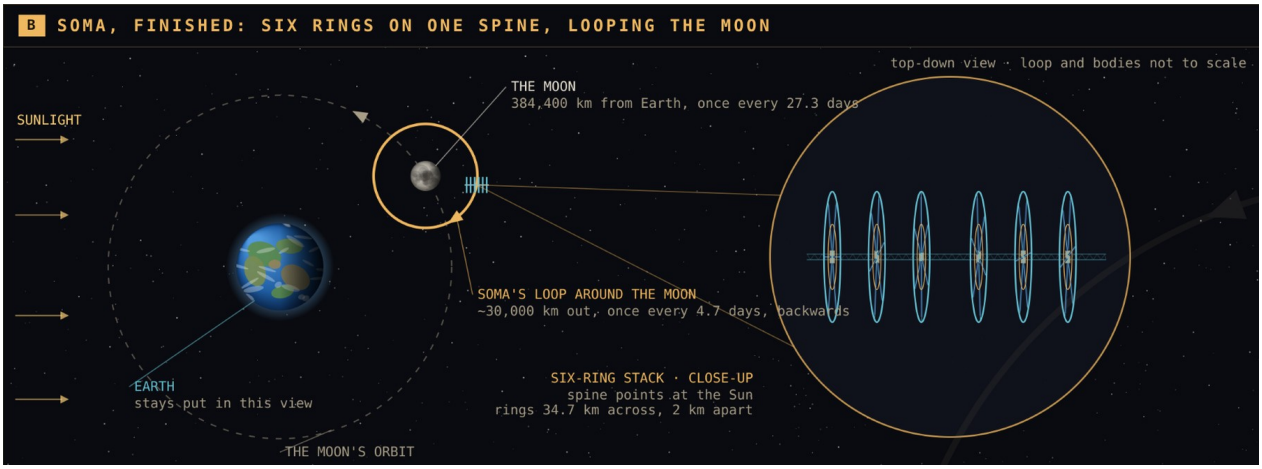


Figure 20: Soma's loop around the Moon, top-down with Earth fixed.

9 Cost per resident

Build cost per resident falls from $168k–661k for ring 1 to $87k–502k for ring 100, because later rings reuse the lunar industry and factories and Earth-made parts get cheaper. The range runs from an optimistic case (80% learning curve) to a conservative one (90%). A standalone ring costs about $89–356B; 100 rings total about $5.2T (optimistic) to $28.7T (conservative).

10 Key numbers

Item	Value
Ring	109 km around, 17.35 km radius, 1 g at 0.23 rpm
Tube	150 m across, 45 levels of 3.33 m, 606 cells of 180 m
Population	∼550,000 per ring; ∼3.3 million per six-ring stack
Atmosphere	57 kPa, 34% O_2
Mass per ring	∼300 Mt, mostly lunar regolith
Spine	280 Starships (14 km core) + 5 drums to $r = 100$ m, ∼1.4 Mt
Middle ring	$r = 8.7$ km, 0.5 g, Ø 40 m
Stack	6 rings, 3 counter-rotating pairs, 2 km apart
Power	∼6 GW per ring from spoke wings
Orbit	Lunar DRO, ∼30,000 km, 4.7-day loop
Cost per resident	Ring 1: $168k–661k → ring 100: $87k–502k

11 Deliverables

Project page: varunpra97.github.io/SomaCislunarSettlement

- **Repository:** github.com/varunpra97/SomaCislunarSettlement, with the README, videos, design boards, logo and render scripts.

- **Full report:** Soma: Cost, Deployment and Operations, 36 pages.
- **Construction film:** *How Soma is built*, 5 min 57 s, from the spine to the zoom out to the orbit.
- **Poster:** 60 × 36 in print PDF, plus a 4096 px image for X.
- **Animated poster:** 8-second looping GIF in which every tile plays its build step, at 3840 px and 1920 px.

12 Caveats and open questions

These are concept estimates; the items below would need real engineering before any of the numbers hold.

- ☐ Basalt-fibre strength and creep at 1.5 GPa over decades of spin
- ☐ How the origami spoke links fold, lock and carry load at 17 km length
- ☐ Long-term orbit stability with a full-physics model, not the planar bicircular one used here
- ☐ Nitrogen supply: lunar N_2 is scarce, so it must come from asteroids or Earth
- ☐ Mass-driver catch accuracy and throughput at ∼30 Mt a year
- ☐ Cost model assumptions: learning rates, Starship launch prices and lunar industry ramp
- ☐ Mass-driver catcher design and how packets are unpacked and placed at Soma
- ☐ Per-cell split of the Earth kit (estimated by function, not itemised)